

- 28 D Let  $p$  be the momentum of the electron.

$$\text{kinetic energy of electron } E = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mE}$$

$$\text{de Broglie wavelength: } \lambda_{\text{electron}} = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$$

$$\text{Energy of a photon } E = \frac{hc}{\lambda_{\text{photon}}} \Rightarrow \lambda_{\text{photon}} = \frac{hc}{E}$$

$$\frac{\lambda_{\text{electron}}}{\lambda_{\text{photon}}} = \frac{h}{\sqrt{2mE}} \times \frac{E}{hc} = \frac{1}{c} \sqrt{\frac{E}{2m}}$$

- 29 D Activity in  $1.00 \text{ m}^3$  of the air:  $0.600 / 4.8 \times 10^{-3} = 125 \text{ Bq}$

$$A = \lambda N \rightarrow N = \frac{A}{\lambda} = \frac{125}{7.55 \times 10^{-3} / 3600} = 5.96 \times 10^7 \text{ Radon atoms}$$

Option A: Students may have thought that the activity is equivalent to the number of radon atoms.

Option B: Students did not convert hour into second for the decay constant.

$$A = \lambda N \rightarrow N = \frac{A}{\lambda} = \frac{125}{7.55 \times 10^{-3}} = 1.66 \times 10^4$$

Option C: Students did not determine the activity in  $1.00 \text{ m}^3$  of the air.

$$A = \lambda N \rightarrow N = \frac{A}{\lambda} = \frac{0.600}{7.55 \times 10^{-3} / 3600} = 2.86 \times 10^5$$

- 30 B  $8.1 \times 10^{-8} \times 1.6 \times 10^{-19} \times 1.0 \times 10^3 = 1.3 \times 10^{-23} \text{ J}$